S181M116 Derivatives and Alternative Investments

Lecture 4: Options, Put-Call Parity, and Binomial Trees

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Today

1. What options are and how they differ from forwards and futures.
2. How calls, puts, long positions, and short positions pay off.
3. How option contracts are specified, traded, margined, cleared, and customized.
4. Which inputs move option prices.
5. How no-arbitrage gives option bounds and put-call parity.
6. When early exercise matters for American options.
7. How binomial trees price European and American options.
8. How delta connects option pricing and hedging.

Core Message

Options are asymmetric contracts

- The **holder** has a right, not an obligation.
- The **writer** receives the premium, but accepts a contingent obligation.
- The premium is paid up front because the payoff is one-sided.

Pricing lesson

We do not price an option by discounting an expected payoff at an arbitrary risky rate. We price it by no-arbitrage relative to the underlying asset and the risk-free asset.

Section 1

Option Contracts and Markets

1. Option Contracts and Markets
2. Option Price Restrictions
3. Early Exercise
4. Binomial Trees
5. Review

Forwards versus Options

Forward or futures contract

- Both parties commit to a future transaction.
- Long and short payoffs are symmetric.
- Initial contract value is often close to zero.
- Margin may be required, but no option premium is paid.

Option contract

- Holder chooses whether to exercise.
- Writer must perform if assigned.
- Payoff is asymmetric.
- Holder pays an option premium up front.

Options are valuable because the holder can keep favorable states and discard unfavorable states.

Calls and Puts

Call option

Right to **buy** the underlying asset at strike K.

$$\text{payoff} = \max(S_T - K, 0).$$

The call holder benefits from higher underlying prices.

Put option

Right to **sell** the underlying asset at strike K.

$$\text{payoff} = \max(K - S_T, 0).$$

The put holder benefits from lower underlying prices.

The strike price is the exercise price. The expiration date is the last date relevant for exercise.

Intuition Example: Insurance

Think of a put option as price insurance.

Investor owns a stock at 100

The investor buys a three-month put with strike 95.

- If the stock falls to 70, the put allows sale at 95.
- If the stock rises to 120, the investor ignores the put and keeps the upside.
- The premium is the cost of this protection.

Teaching point

The put does not prevent the stock price from falling. It changes the investor's payoff after the fall.

Intuition Example: Reservation Price

Think of a call option as locking in a maximum purchase price.

Firm may need copper in three months

The firm buys a call with strike 9,000 per ton.

- If copper rises to 10,500, the call lets the firm buy at 9,000.
- If copper falls to 8,000, the firm buys in the market and lets the call expire.
- The premium pays for flexibility.

Comparison with a forward

A forward would lock in the purchase price both when copper rises and when copper falls.

European and American Exercise

European option

Can be exercised only at maturity.

American option

Can be exercised at any time up to and including maturity.

These names describe exercise style, not geography.

Value implication

American exercise cannot make the holder worse off:

$$C \geq c, \qquad P \geq p.$$

Here C, P are American prices and c, p are European prices.

Payoff versus Profit

Payoff

Cash value received at maturity before subtracting the premium.

Profit

Payoff minus the initial premium, usually ignoring time value of money in diagrams.

Call with $K = 100$ and premium 5

If $S_T = 102$, the holder exercises:

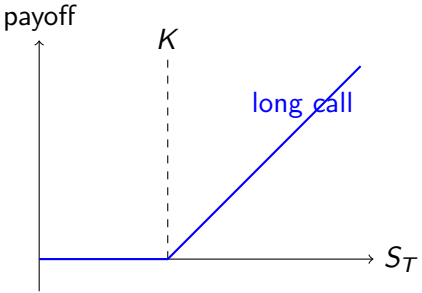
$$\text{payoff} = 2, \qquad \text{profit} = 2 - 5 = -3.$$

Exercising still reduces the loss relative to letting the option expire worthless.

Long Call Payoff

$$\text{payoff} = \max(S_T - K, 0)$$

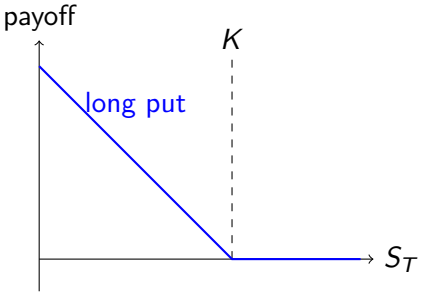
- Limited downside: premium paid.
- Potentially unlimited upside for a stock.
- Break-even at $S_T = K + \text{premium}$.



Long Put Payoff

$$\text{payoff} = \max(K - S_T, 0)$$

- Limited downside: premium paid.
- Payoff rises as the asset falls.
- Break-even at $S_T = K - \text{premium}$.



The Four Basic Positions

Position	Payoff at maturity	Risk profile
Long call	$\max(S_T - K, 0)$	Pays premium; benefits from upside
Short call	$-\max(S_T - K, 0)$	Receives premium; loses on upside
Long put	$\max(K - S_T, 0)$	Pays premium; benefits from downside
Short put	$-\max(K - S_T, 0)$	Receives premium; loses on downside

Writer risk

Short option positions receive cash today but create possible future liabilities. This is why option writers face margin requirements.

Example: Covered Call versus Naked Call

Covered call

Investor owns 100 shares and writes one call.

- Premium provides income.
- Upside above the strike is given away.
- If assigned, the investor delivers shares already owned.

Naked short call

Investor writes the call without owning shares.

- Premium is received up front.
- If the stock rises sharply, losses can be very large.
- The writer must buy shares at the market price to deliver at K.

Moneyness

	Call	Put
In the money	$S > K$	$S < K$
At the money	$S = K$	$S = K$
Out of the money	$S < K$	$S > K$

Intrinsic value

$$\text{Call intrinsic value} = \max(S - K, 0), \qquad \text{Put intrinsic value} = \max(K - S, 0).$$

Time value

$$\text{Time value} = \text{option price} - \text{intrinsic value}.$$

Example: Reading an Option Quote

A stock trades at 48. A quoted option is:

$$\text{XYZ 50 call, two months, price 2.40.}$$

- It is a call, so it gives the right to buy.
- The strike is 50.
- It is out of the money because $48 < 50$.
- Intrinsic value is $\max(48 - 50, 0) = 0$.
- The full 2.40 price is time value.
- One standard contract costs $2.40 \times 100 = 240$.

Question for students

What stock price at expiration is needed to break even, ignoring interest?

Stock Option Contract Terms

- A standard U.S. exchange-traded stock option contract usually covers 100 shares.
- A quoted price is typically per share.
- A quote of 4.20 means a premium of $4.20 \times 100 = 420$ for one contract.
- Exchanges specify expiration dates, strikes, exercise style, contract size, and adjustment rules.
- Weekly options and longer-term LEAPS add maturity choices beyond standard monthly cycles.

Class versus series

Option class: all calls or all puts on the same underlying.
Option series: same type, same underlying, same strike, and same expiration.

Dividends, Splits, and Adjustments

Cash dividends

Ordinary cash dividends usually do not mechanically adjust exchange-traded stock option terms. Expected dividends enter option prices because they affect the stock price path.

Stock splits and stock dividends

Contract terms are adjusted to preserve economic exposure.

Two-for-one split

A call on 100 shares with strike 30 becomes a call on 200 shares with strike 15.

$$100 \times 30 = 200 \times 15 = 3000.$$

Trading, Clearing, and Margin

Market makers

Quote bid and ask prices, provide liquidity, earn bid-ask spreads, and hedge inventory risk.

Offsetting orders

Most positions are closed by trading the same option series in the opposite direction.

Margin

Buyers pay the premium and have no further obligation. Writers have contingent obligations and must post margin.

OCC

The Options Clearing Corporation records positions, handles exercise, and guarantees performance.

Exchange-Traded and OTC Options

Exchange-traded

- Standardized contract terms.
- Central clearing.
- Transparent listed markets.
- Lower counterparty risk.

Over the counter

- Customized strikes, maturities, payoffs, and settlement.
- Useful for institutional hedging.
- Exposes parties to credit, collateral, model, and liquidity risk.

Related contracts include warrants, employee stock options, and convertible bonds.

Section 2

Option Price Restrictions

1. Option Contracts and Markets
2. Option Price Restrictions
3. Early Exercise
4. Binomial Trees
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Six Inputs to a Stock Option Price

1. Current stock price S_0.
2. Strike price K.
3. Time to expiration T.
4. Volatility σ.
5. Risk-free interest rate r.
6. Expected dividends during the option's life.

Increase in input	Call value	Put value
S_0	Increases	Decreases
K	Decreases	Increases
σ	Increases	Increases
r	Usually increases	Usually decreases
Expected dividends	Decreases	Increases

Why Volatility Raises Both Calls and Puts

The underlying owner faces both sides

A stock investor gains from upside and loses from downside.

The option holder keeps only one side

- A call keeps upside above K and discards downside below K.
- A put keeps downside below K and discards upside above K.

Convex payoff

Higher uncertainty makes convex payoffs more valuable, holding the current underlying price fixed.

Intuition Example: Same Average, Different Volatility

Consider a call with strike 100.

Low-volatility world

Stock ends at:

$$98 \quad \text{or} \quad 102.$$

Call payoff:

$$0 \quad \text{or} \quad 2.$$

High-volatility world

Stock ends at:

$$80 \quad \text{or} \quad 120.$$

Call payoff:

$$0 \quad \text{or} \quad 20.$$

Both worlds can have the same average stock price, but the call is much more valuable in the high-volatility world.

Upper Bounds

Calls

A call cannot be worth more than the stock:

$$c \leq S_0, \qquad C \leq S_0.$$

Puts

A put cannot be worth more than the strike:

$$p \leq K, \qquad P \leq K.$$

For a European put, a sharper bound is

$$p \leq Ke^{-rT}.$$

Lower Bound for a European Call

For a non-dividend-paying stock:

$$c \geq \max(S_0 - Ke^{-rT}, 0).$$

Dominance argument

Compare:

- Portfolio A: one European call plus cash Ke^{-rT};
- Portfolio B: one share.

At maturity, Portfolio A is worth at least the stock in every state.

Example

If $S_0 = 20$, $K = 18$, $r = 10\%$, $T = 1$:

$$c \geq 20 - 18e^{-0.10} = 3.71.$$

Lower Bound for a European Put

For a non-dividend-paying stock:

$$p \geq \max(Ke^{-rT} - S_0, 0).$$

Dominance argument

Compare:

- Portfolio A: one European put plus one share;
- Portfolio B: cash Ke^{-rT}.

At maturity, Portfolio A is worth at least K in every state.

Example

If $S_0 = 37$, $K = 40$, $r = 5\%$, $T = 0.5$:

$$p \geq 40e^{-0.025} - 37 = 2.01.$$

Put-Call Parity

For European options on a non-dividend-paying stock:

$$c + Ke^{-rT} = p + S_0.$$

Fiduciary call

$$\text{call} + \text{cash } Ke^{-rT}$$

Pays $\max(S_T, K)$ at maturity.

Protective put

$$\text{put} + \text{stock}$$

Pays $\max(S_T, K)$ at maturity.

Identical future payoffs must have identical current values.

Parity State-by-State

State at T	Call + cash	Put + stock	Common payoff
$S_T > K$	$S_T - K + K$	$0 + S_T$	S_T
$S_T \leq K$	$0 + K$	$K - S_T + S_T$	K

Therefore:

$$c - p = S_0 - Ke^{-rT}.$$

Interpretation

The call-put price difference equals the forward-like value of buying the stock and financing the strike.

Intuition Example: Two Ways to Guarantee a Floor

Suppose $K = 100$. There are two ways to guarantee that the final portfolio is worth at least 100.

Protective put

Buy the stock and buy a put.

$$S_T + \max(100 - S_T, 0) = \max(S_T, 100).$$

Fiduciary call

Buy a call and lend the present value of 100.

$$\max(S_T - 100, 0) + 100 = \max(S_T, 100).$$

Parity intuition

If two strategies deliver the same insured payoff, they must cost the same today.

Parity Example

Suppose

$$S_0 = 31, \qquad K = 30, \qquad r = 6\%, \qquad T = 0.25, \qquad c = 3.$$

Put-call parity implies

$$p = c + Ke^{-rT} - S_0.$$

Hence

$$p = 3 + 30e^{-0.06(0.25)} - 31 \approx 1.55.$$

Parity Arbitrage

If

$$c + Ke^{-rT} > p + S_0,$$

the fiduciary call is expensive. Sell it and buy the protective put:

$$\text{short call, borrow } Ke^{-rT}, \text{ buy put, buy stock.}$$

If

$$c + Ke^{-rT} < p + S_0,$$

the protective put is expensive. Buy the fiduciary call and sell the protective put:

$$\text{buy call, lend } Ke^{-rT}, \text{ short put, short stock.}$$

Reality check

Transaction costs, bid-ask spreads, short-sale constraints, and funding spreads determine whether the arbitrage is implementable.

Dividends and Put-Call Parity

Let D be the present value of known cash dividends during the option's life.

Dividend-adjusted parity

$$c + D + Ke^{-rT} = p + S_0.$$

Equivalently:

$$c - p = S_0 - D - Ke^{-rT}.$$

Lower bounds

$$c \geq \max(S_0 - D - Ke^{-rT}, 0),$$

$$p \geq \max(Ke^{-rT} + D - S_0, 0).$$

American Options and Parity Inequalities

Exact European put-call parity generally does not hold for American options because early exercise can matter.

Non-dividend-paying stock

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}.$$

- American calls on non-dividend-paying stocks have no early-exercise premium.
- American puts may have an early-exercise premium.
- Therefore American parity becomes an inequality, not an equality.

Section 3

Early Exercise

1. Option Contracts and Markets
2. Option Price Restrictions
3. Early Exercise
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American Calls Without Dividends

Result

It is never optimal to exercise an American call early on a non-dividend-paying stock.

Exercising early gives up:

1. the time value of the option;
2. the ability to delay paying the strike K;
3. downside protection if the stock later falls.

Therefore:

$$C = c$$

for an otherwise identical non-dividend-paying stock call.

Example: Why Not Exercise a Call Early?

You hold an American call on a non-dividend-paying stock:

$$S = 120, \qquad K = 100, \qquad \text{six months remaining.}$$

Exercising now gives intrinsic value:

$$120 - 100 = 20.$$

But selling the call is usually better because the call price includes:

- intrinsic value;
- remaining time value;
- the benefit of not paying K until later.

Classroom takeaway

If you want to exit, sell the call. Do not throw away time value by exercising early.

American Puts Without Dividends

Result

Early exercise of an American put can be optimal.

- A deep-in-the-money put allows the holder to sell the stock for K.
- If the stock price is very low, immediate exercise produces cash early.
- Receiving K early is valuable when interest rates are positive.

The American put must be worth at least intrinsic value:

$$P \geq \max(K - S_0, 0).$$

Dividends and Early Exercise

Calls

Dividends can make early exercise optimal. A call holder does not receive dividends unless the option is exercised and converted into stock.

Puts

Dividends reduce the stock price when the stock goes ex-dividend, which can make waiting more attractive for puts.

Practical rule

Early exercise depends on moneyness, interest rates, dividend timing, dividend size, and remaining time value.

Section 4

Binomial Trees

1. Option Contracts and Markets
2. Option Price Restrictions
3. Early Exercise
4. **Binomial Trees**
5. Review

One-Step Binomial Setup

Over one period, the stock moves from S_0 to one of two values:

$$S_u = S_0 u, \qquad S_d = S_0 d, \qquad u > 1, \quad d < 1.$$

$$\begin{tikzcd}
 & S_0 u \\
S_0 \arrow[ur, "\text{up}"] \arrow[dr, "\text{down}"'] & \\
 & S_0 d
\end{tikzcd}$$

Let the option values at the two final nodes be f_u and f_d.

Intuition Example: A Tiny Replication Problem

Suppose a stock is 20 today and next period is either 22 or 18. A call with strike 21 pays:

$$1 \quad \text{if stock is } 22, \qquad 0 \quad \text{if stock is } 18.$$

Question

Can we combine stock and cash to reproduce exactly the same payoff?

- If yes, the option must cost the same as the replicating portfolio.
- This is the economic content of binomial option pricing.

Riskless Portfolio

Build a portfolio:

$$\Delta \text{ shares of stock} - 1 \text{ option.}$$

Up-state value:

$$S_0 u \Delta - f_u.$$

Down-state value:

$$S_0 d \Delta - f_d.$$

Choose Δ so the two values are equal:

$$S_0 u \Delta - f_u = S_0 d \Delta - f_d.$$

Therefore:

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}.$$

Risk-Neutral Valuation

The one-step option value can be written as:

$$f = e^{-rT}\left[pf_u + (1-p)f_d\right],$$

where

$$p = \frac{e^{rT} - d}{u - d}.$$

Meaning of p

p is the probability that makes the stock earn the risk-free rate:

$$pS_0u + (1-p)S_0d = S_0e^{rT}.$$

Do not confuse probabilities

p is not generally the real-world probability of an up move. It is the risk-neutral probability used for pricing.

One-Step Call Example

Suppose

$$S_0 = 20, \quad u = 1.10, \quad d = 0.90, \quad K = 21, \quad r = 12\%, \quad T = 0.25.$$

Then

$$S_u = 22, \qquad S_d = 18.$$

For a European call:

$$f_u = \max(22 - 21, 0) = 1, \qquad f_d = \max(18 - 21, 0) = 0.$$

Risk-neutral probability:

$$p = \frac{e^{0.12(0.25)} - 0.90}{1.10 - 0.90} \approx 0.6523.$$

Price:

$$f = e^{-0.12(0.25)}[0.6523(1) + 0.3477(0)] \approx 0.633.$$

Delta in the Example

In the one-step call example:

$$\Delta = \frac{f_u - f_d}{S_u - S_d} = \frac{1 - 0}{22 - 18} = 0.25.$$

Hedging interpretation

A trader short one call buys 0.25 shares to create a one-step riskless hedge.

Dynamic point

In a multi-step tree, delta changes by node. A hedge must be rebalanced as the stock price and time to maturity change.

Example: Delta as Shares per Option

At a node, a call is worth 9 if the stock goes to 66 and 1 if the stock goes to 54.

$$\Delta = \frac{9-1}{66-54} = \frac{8}{12} = 0.6667.$$

Trader short 100 calls

The one-step delta hedge buys approximately

$$100 \times 0.6667 = 66.67$$

shares.

Interpretation

Delta is not a permanent hedge ratio. It is local to the current node and time step.

Why Expected Return Does Not Enter

The tempting but wrong instinct

"The option is risky, so discount its expected payoff at an option-specific risky discount rate."

No-arbitrage logic

The option is priced relative to the stock. The stock price S_0 already reflects risk preferences and required expected return.

Key lesson

Given S_0, u, d, r, and the payoff, the option price is pinned down by arbitrage. The stock's real-world expected return μ drops out.

Two-Step Trees

Starting from S_0, two steps give terminal stock prices:

$$S_0 u^2, \qquad S_0 ud, \qquad S_0 d^2.$$

$$\begin{tikzcd}
 & & S_0 u^2 \\
 & S_0 u \arrow[ur] \arrow[dr] & \\
S_0 \arrow[ur] \arrow[dr] & & S_0 ud \\
 & S_0 d \arrow[ur] \arrow[dr] & \\
 & & S_0 d^2
\end{tikzcd}$$

Recombining trees keep the number of nodes manageable.

Backward Induction

1. Compute terminal option payoffs.
2. Step backward one period at a time.
3. At each node, use

$$f = e^{-r\Delta t}\left[pf_u + (1-p)f_d\right].$$

For two steps:

$$f_u = e^{-r\Delta t}\left[pf_{uu} + (1-p)f_{ud}\right],$$

$$f_d = e^{-r\Delta t}\left[pf_{ud} + (1-p)f_{dd}\right],$$

$$f = e^{-r\Delta t}\left[pf_u + (1-p)f_d\right].$$

Two-Step European Put Example

Consider a two-year European put:

$$S_0 = 50, \quad K = 52, \quad u = 1.2, \quad d = 0.8, \quad r = 5\%, \quad \Delta t = 1.$$

Risk-neutral probability:

$$p = \frac{e^{0.05} - 0.8}{1.2 - 0.8} \approx 0.6282.$$

Terminal stock prices:

$$72, \quad 48, \quad 32.$$

Terminal put payoffs:

$$f_{uu} = 0, \quad f_{ud} = 4, \quad f_{dd} = 20.$$

Price:

$$f = e^{-0.10}\left[(0.6282)^2(0) + 2(0.6282)(0.3718)(4) + (0.3718)^2(20)\right] \approx 4.19.$$

American Option Trees

For American options, every node has an exercise decision.

Continuation value

$$e^{-r\Delta t}\left[pf_u + (1-p)f_d\right].$$

American put node value

$$\max\left[K - S,\ e^{-r\Delta t}\left(pf_u + (1-p)f_d\right)\right].$$

Algorithm

Work backward. At each node, compute continuation value, compute exercise value, and keep the larger.

American Put Node Example

At a node in an American put tree:

$$S = 38, \quad K = 45, \quad r = 4\%, \quad \Delta t = 0.5, \quad p = 0.55,$$

$$f_u = 5, \qquad f_d = 12.$$

Continuation value:

$$e^{-0.04(0.5)}[0.55(5) + 0.45(12)] = e^{-0.02}(8.15) \approx 7.99.$$

Exercise value:

$$K - S = 45 - 38 = 7.$$

Node value is 7.99, so the put is not exercised at this node.

Choosing u, d, and p

A tree with time step Δt is matched to volatility by setting

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}.$$

Once u and d are chosen, no-arbitrage fixes

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

Measure change intuition

Moving from the real-world measure to the risk-neutral measure changes expected returns, not the volatility used to build the tree.

Tree Parameter Example

Suppose

$$\sigma = 30\%, \qquad r = 5\%, \qquad \Delta t = 1.$$

Then

$$u = e^{0.30} = 1.3499, \qquad d = e^{-0.30} = 0.7408.$$

The risk-neutral probability is

$$p = \frac{e^{0.05} - 0.7408}{1.3499 - 0.7408} \approx 0.5097.$$

These parameters define the tree. Valuation then proceeds by backward induction.

Increasing the Number of Steps

If option life is T and there are n steps:

$$\Delta t = \frac{T}{n}.$$

Use

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}}, \qquad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Convergence

As n increases, the time grid becomes finer. For European options on non-dividend-paying stocks, the binomial price converges to the Black-Scholes-Merton price.

Five-Step Illustration

Suppose

$$T = 2, \quad n = 5, \quad r = 5\%, \quad \sigma = 30\%.$$

Then

$$\Delta t = \frac{2}{5} = 0.4.$$

$$u = e^{0.30\sqrt{0.4}} \approx 1.2089, \qquad d = \frac{1}{u} \approx 0.8272.$$

$$e^{r\Delta t} = e^{0.05(0.4)} \approx 1.0202,$$

$$p = \frac{1.0202 - 0.8272}{1.2089 - 0.8272} \approx 0.5056.$$

Section 5

Review

1. Option Contracts and Markets

2. Option Price Restrictions

3. Early Exercise

4. Binomial Trees

5. Review

What We Omitted

Intentional exclusions

The source material includes:

- Section 13.10 on DerivaGem software;
- Section 13.11 on options on other assets.

These slides follow the lecture notes and exclude those sections. The binomial-tree discussion here is limited to stock options through increasing the number of steps.

Checklist

After this lecture you should be able to:

- compute payoffs and profits for long and short calls and puts;
- interpret moneyness, intrinsic value, and time value;
- explain why option writers post margin;
- list the inputs that affect call and put prices;
- derive option upper and lower bounds;
- use put-call parity and identify parity arbitrage;
- explain early exercise for American calls and puts;
- price one-step and two-step binomial trees;
- compute delta and interpret the hedge;
- choose u, d, and p from volatility and the time step.

Practice Problems

1. A call has strike 60 and premium 4. Compute payoff and profit for $S_T = 52, 60, 71$.
2. A put has strike 45 and premium 3.50. Compute payoff and profit for $S_T = 35, 45, 52$.
3. A European call costs 6.20. The stock is 50, the strike is 52, $r = 5\%$, and $T = 0.75$. Use put-call parity to compute the put price.
4. A stock is 30 and can move to 36 or 24 in one year. A one-year call has strike 32 and $r = 5\%$. Compute p, the call price, and delta.
5. At an American put node, $S = 42$, $K = 50$, $r = 6\%$, $\Delta t = 0.25$, $p = 0.52$, $f_u = 5$, and $f_d = 11$. Should the option be exercised?

Next Lecture

Option trading strategies and payoff engineering